

§3.6 Implicit Differentiation

Homework: 5, 9, 15, 21, 25, 36, 41, 43, 54, 67

Question : 一個二元方程式是否隱藏函數的關係？

例如 : (i) $x + y = 1$ 能否寫成

$$x + (f(x)) = 1, \text{ where } y = f(x)?$$

(ii) $x^2 + y^2 = 1$ 能否寫成

$$x^2 + (f(x))^2 = 1, \text{ where } y = f(x)?$$

(iii) $x^3 + y^3 = 6xy$ 能否寫成

$$x^3 + (f(x))^3 = 6x(f(x)), \text{ where } y = f(x)?$$

(iv) 給一個二元方程式 $g(x, y) = 0$ 能否將此方程式寫成

$$g(x, f(x)) = 0?$$

很明顯 (i) 的答案是肯定的，(ii)、(iii)、(iv) 的答案是較複雜。數學系大二的高微課中的隱函數的定理 (Implicit Function Theory) 會給出較完整的答案。粗略的講，除了少數的點之外，其它點的局部附近上述問題的答案是肯定的。在初微的課程中，我們因此都假設：The given equation determines y (respectively, x) implicitly as a differentiable function of x (respectively, y) so that the method of implicit differentiation can be applied .

Example1 : $x^2 + y^2 = 25$, 求 $\frac{dy}{dx}, \frac{dx}{dy}$ and equation of the tangent at (3,4) .

$$\begin{array}{l} \text{Solution :} \quad 2x + 2y \left(\frac{dy}{dx} \right) = 0 \quad \left| \quad 2x \left(\frac{dx}{dy} \right) + 2y = 0 \right. \\ \Rightarrow \quad \frac{dy}{dx} = -\frac{x}{y} . \quad \left| \quad \frac{dx}{dy} = -\frac{y}{x} . \right. \\ m = \frac{dy}{dx} \Big|_{(3,4)} = -\frac{3}{4} . \\ (y-4) = -\frac{3}{4}(x-3) . \end{array}$$

Example2 : $x^3 + y^3 = 6xy$ 求 $\frac{dy}{dx} = ?$

$$\begin{array}{l} \text{Solution :} \quad 3x^2 + 3y^2 \left(\frac{dy}{dx} \right) = 6y + 6x \left(\frac{dy}{dx} \right) \\ \frac{dy}{dx} = \frac{2y - x^2}{y^2 - 2x} . \end{array}$$

Example3 : $\sin(x+y) = y^2 \cos x$ 求 $\frac{dy}{dx} = ?$

$$\begin{array}{l} \text{Solution :} \quad \cos(x+y) \left(1 + \frac{dy}{dx} \right) = 2y \left(\frac{dy}{dx} \right) \cos x - y^2 \sin x \\ \frac{dy}{dx} = \frac{\cos(x+y) + y^2 \sin x}{2y \cos x - \cos(x+y)} . \end{array}$$

Example4 : $f(f^{-1}(x)) = x$.

(i) Find $(f^{-1})'(x)$.

(ii) if $f(4) = 5$, $f'(4) = \frac{2}{3}$, 求 $(f^{-1})'(5) = ?$

Solution : (i) $f'(f^{-1}(x))(f^{-1})'(x) = 1$

$$\Rightarrow (f^{-1})'(x) = \frac{1}{f'(f^{-1}(x))} .$$

$$(ii) \left(f^{-1}\right)'(5) = \frac{1}{f'(4)} = \frac{3}{2}.$$

反三角函數的微分

$$(i) \left(\sin^{-1} x\right)' = \frac{1}{\sqrt{1-x^2}}$$

$$(ii) \left(\cos^{-1} x\right)' = \frac{-1}{\sqrt{1-x^2}}$$

$$(iii) \left(\tan^{-1} x\right)' = \frac{1}{1+x^2}$$

$$(iv) \left(\cot^{-1} x\right)' = \frac{-1}{1+x^2}$$

$$(v) \left(\sec^{-1} x\right)' = \frac{1}{x\sqrt{x^2-1}}$$

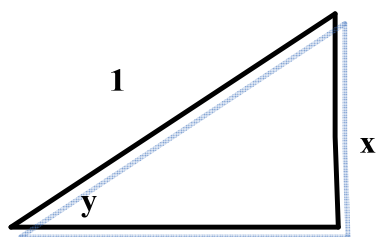
$$(vi) \left(\csc^{-1} x\right)' = \frac{-1}{x\sqrt{x^2-1}}$$

($\csc^{-1} x, \sec^{-1} x \in \text{III 象限}$ when $x < 0$)

$$1. \quad y = \sin^{-1} x \quad \Rightarrow \quad x = \sin y$$

$$1 = \cos y \left(\frac{dy}{dx} \right)$$

$$\frac{dy}{dx} = \frac{1}{\cos y} = \frac{1}{\sqrt{1-x^2}}.$$



$$2. \quad y = \cos^{-1} x$$

(Method 1) :

$$\Rightarrow \quad x = \cos y$$

$$1 = -\sin y \left(\frac{dy}{dx} \right)$$

$$\frac{dy}{dx} = -\frac{1}{\sin y} = \frac{-1}{\sqrt{1-x^2}}.$$

(Method 2) :

$$\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}$$

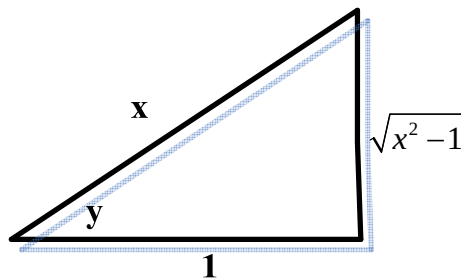
$$\Rightarrow (\sin^{-1} x)' + (\cos^{-1} x)' = 0$$

$$\Rightarrow (\cos^{-1} x)' = -(\sin^{-1} x)' = \frac{-1}{\sqrt{1-x^2}} .$$

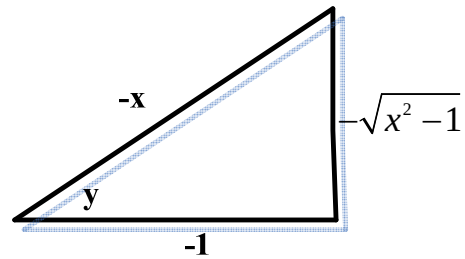
$$3. \quad y = \sec^{-1} x \quad \Rightarrow \quad x = \sec y$$

$$1 = \sec y \tan y \frac{dy}{dx}$$

$$\frac{dy}{dx} = \frac{1}{\sec y \tan y} = \frac{1}{x\sqrt{x^2-1}}$$



$$(i) \quad x > 0, 0 \leq y < \frac{\pi}{2}.$$



$$(ii) \quad x < 0 (\Rightarrow -x > 0), \pi \leq y < \frac{3\pi}{2}.$$

Note that if $\sec^{-1} x \in \text{second quadrant}$ when $x < 0$,

$$\text{Then } \frac{dy}{dx} = \begin{cases} \frac{1}{x\sqrt{x^2-1}} & x > 0 \\ \frac{1}{-x\sqrt{x^2-1}} & x < 0. \end{cases}$$

That is why in this book, $\sec^{-1} x$ is chosen to be in the third quadrant

whenever $x < 0$.